# Sajid Muhaimin Choudhury, Ph.D.

Professor, Department of Electrical and Electronic Engineering, BUET, Dhaka

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## Profile

Professor in the Department of Electrical and Electronic Engineering at the Bangladesh University of Engineering and Technology (BUET); Ph.D., Purdue University. Research spans quantum computing, photonics, antennas, computing, embedded systems and renewable energy. Senior Member of IEEE, Founding President of the Optica Bangladesh Section, and Founding Chair of the IEEE Photonics Society Bangladesh Chapter.

## Appointments

**Bangladesh University of Engineering and Technology (BUET), Dhaka**

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| Jul 2025 – present | **Professor**, Department of Electrical and Electronic Engineering |
| Jul 2022 – Jul 2025 | **Associate Professor**, Department of Electrical and Electronic Engineering |
| Jun 2013 – Jul 2022 | **Assistant Professor**, Department of Electrical and Electronic Engineering |
| Jan 2010 – Jun 2013 | **Lecturer**, Department of Electrical and Electronic Engineering |
| Nov 2009 – Jan 2010 | **Lecturer**, Institute of Information and Communication Technology (IICT) |

## Education

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|---|---|
| Aug 2013 – Aug 2019 | **Ph.D.**, School of Electrical and Computer Engineering, Purdue University, West Lafayette, IN, USA<br>Thesis: *Wavefront Manipulation with Metasurfaces Based on New Materials.* Co-supervisors: Alexandra Boltasseva and Alexander V. Kildishev. |
| Aug 2011 – 2013 | **M.Sc. Engg.**, Department of Electrical and Electronic Engineering, BUET<br>Thesis: *Design of a Fractal Antenna Based on Hexaflake Fractal Structure.* Supervisor: Dr. M. A. Matin. |
| Dec 2004 – Aug 2010 | **B.Sc. Engg.**, Department of Electrical and Electronic Engineering, BUET<br>CGPA 3.94/4.00. Thesis: *Design and Analysis of a Multiband Dual Feed Axially Symmetric Cassegrain Antenna System.* Supervisor: Dr. M. A. Matin. |
| 2004 | **H.S.C.**, Notre Dame College, Dhaka GPA 5.00/5.00 |
| 2002 | **S.S.C.**, Udayan Uchchya Madhyamic Bidyalaya, Dhaka GPA 5.00/5.00 |

## Research

Research is organised under the **Q-PACERS** framework: Quantum, Photonic, Antenna, Computing, Embedded and Renewable-energy Systems, within the Electronics and Photonics (EP) division of EEE.

**Nanophotonics and plasmonics.** The group develops photonic materials and devices at the nanoscale, designing plasmonic nanostructures for optical sensing and modulation. A recent publication reports a concentric plasmonic resonator for dual-band absorption in the near- and mid-infrared, aimed at biochemical sensing. Plasmonic biosensors and high-speed electro-optic modulators are continuing interests, with

the goal of nanophotonic systems that extend light–matter interaction for communications, healthcare and computing.

**Advanced antennas and metasurfaces.** Building on a background in antenna engineering (his M.Sc. research produced a microstrip patch antenna with a hexaflake fractal geometry), the group studies metasurfaces, engineered electromagnetic surfaces, for new applications. A recent thesis student developed an electrically reconfigurable metasurface for metalensing in the near-infrared band. Metasurface-based antennas and optical elements are central to this programme, with applications from wireless communication devices to compact imaging and holography.

**Quantum computing and quantum photonics.** The research extends to quantum information technology: developing and optimising quantum algorithms and exploring practical quantum hardware, in particular photonic systems (photons in nanoscale devices and integrated photonic circuits) that realise qubits and quantum logic. A dedicated graduate course on quantum computing and photonic technologies (see Teaching) lays the groundwork for a research programme in quantum algorithms and photonic quantum hardware, contributing to Bangladesh's entry into quantum technology research.

**Embedded systems and computing for IoT.** Design projects pursue efficient computing systems and embedded platforms for the Internet of Things. By integrating sensors, processors and communication interfaces, these platforms are tailored to environmental monitoring, smart grids and healthcare devices. The work complements EEE 416 (Microprocessor and Embedded Systems Design Laboratory), where undergraduate teams prototype IoT systems that contribute to edge-computing and distributed-sensor-network research.

**Renewable-energy and sustainable systems.** The group works on efficient solar-cell design and grid optimisation, and on improving photovoltaic devices through nanotechnology. A recent journal paper demonstrates an efficiency improvement in a crystalline-silicon/thin-film solar cell using hybrid metal–dielectric nanostructures. Photocatalytic water splitting for hydrogen production is a further interest. The plan is to integrate these strands into smart renewable-energy systems, for instance solar converters with embedded IoT sensors for performance optimisation, or photonic devices that improve energy generation and storage.

## Research Grants

Apr 2024 – Oct 2025 **Synergizing Deep Learning and Topology Optimization for Tunable Meta-lens Design**
Principal Investigator. Internal Research Grant, BUET. BDT 14,00,000.

Jan 2023 – Jan 2024 **Structurally Tunable Gear-Shaped Plasmonic Sensor**
Principal Investigator. Basic Research Grant, Research and Innovation Centre for Science and Engineering (RISE), BUET. BDT 7,00,000.

Jan 2021 – Jan 2022 **Design and Low-cost Fabrication of Optical Biosensors for Virus Detection**
Principal Investigator. Basic Research Grant, Center for Advanced Study and Research (CASR), BUET. BDT 1,00,000.

## Teaching and Curriculum Development

Course content is updated continually to reflect the state of the art, with an emphasis on hands-on learning. Beyond delivering core EEE courses, the teaching programme introduces new courses and pedagogical approaches that prepare students for rapidly evolving fields.

EEE 415/416

**Modernisation of the Microprocessor and Embedded Systems curriculum.** EEE 415 (Microprocessors and Embedded Systems) traditionally taught the legacy Intel 8086 microarchitecture and its assembly language, at BUET and at most public and private universities in Bangladesh, whereas comparable courses abroad use modern architectures (MIPS, ARM, RISC-V). He led an overhaul that replaced the 8086-centred syllabus with one built on the ARM architecture and microcontroller design. The revised course covers modern processors and microcontrollers, the ARM Cortex-A and Cortex-M architectures, and high-level design with Verilog. Students learn assembly programming and system design on ARM/RISC-V platforms (stack operations, interrupts, serial interfaces), aligning the curriculum with global standards and the needs of Bangladesh's technology sector.

Lab, 2022

**Modernisation of the Microprocessor and Embedded Systems Laboratory** (lab-in-charge). To support the revised syllabus, he submitted a proposal to the Head of the Department, which was forwarded to and approved by the University Syndicate in January 2022.

- **Pedagogical layout.** Three zones (Class Space, Project Space and Storage Space) for unobstructed instruction, team prototyping benches and organised component inventory. Each workstation, shared by two students, integrates an FPGA board, an MCU kit and a development PC.
- **Core hardware.** STM32 Nucleo ARM Cortex-M4 evaluation kits and Digilent Nexys A7-100T (Artix-7) FPGA trainers anchor the digital-design workflow, supported by 150 MHz digital storage oscilloscopes, programmable power supplies, logic analysers, and sensors and actuators (stepper and servo motors, LCDs, temperature sensors).
- **Rapid prototyping.** A 3D printer for enclosures and a soldering station for PCB prototyping.
- **Budget.** BDT 62.42 lakh in total, with components chosen for low cost and local availability: equipment 26%, test and measurement 36%, IT and furniture 24%, civil and electrical works 14%; 17 complete workstations and specialised project equipment.

EEE 416

**Project-based laboratory.** He redesigned the EEE 416 laboratory around experiential learning in two phases: guided experiments that reinforce fundamentals, followed by open-ended design projects on EEE 415 topics. Student teams build, for example, an embedded IoT device or a small computer system, applying processor architecture, interfacing and programming, and developing problem-solving and hardware-debugging skills in line with Outcome-Based Education principles. The department later adopted the model for most compulsory sessional courses, replacing the capstone project with "Culmination of Program Outcomes with Sessional Course Projects".

EEE 6408, 6505

**New postgraduate courses:** EEE 6408 Nano Systems and EEE 6505 Nanophotonics and Plasmonics, both offered for the first time in the department.

EEE 6002, 2024 — **Quantum Computing and Quantum Photonics** (special topics). First offered in 2024, the course gives EEE graduate students a foundation in quantum information science, from qubits and quantum logic gates to quantum algorithms (Deutsch–Jozsa, Grover, Shor) and error correction, with an overview of current quantum hardware platforms. It also integrates photonics, covering photonic quantum computers and continuous-variable quantum computing, filling a gap in the curriculum and preparing students and faculty for research in quantum technologies. Further offerings are planned, potentially including undergraduate electives in quantum technology, nanophotonics or advanced embedded systems.

EEE 6516 — **Quantum Computing** (3.0 credit hours, EP division), designed from the special-topics course and approved by the Academic Council.

Syllabus: introduction to quantum computing; postulates of quantum mechanics; qubits and single-qubit gates; operators and quantum measurement; quantum information processing, superposition, entanglement, the no-cloning theorem and quantum teleportation; quantum circuits, controlled operations and measurement, universal quantum gates; quantum algorithms: Deutsch–Jozsa, Grover's and Shor's algorithms, the quantum Fourier transform and phase estimation; density matrices and the Bloch-sphere representation; quantum error correction and fault-tolerant architecture; physical realisations: harmonic oscillator, optical quantum computers, trapped ions, superconducting qubits and new implementation schemes.

## Postgraduate Supervision

### Completed

Sep 2025 — Puja Das, M.Sc. Engg.
*Efficiency Enhancement of HTL-Free Perovskite/Cadmium-Sulfide Heterojunction Solar Cell Incorporating Light Trapping Structure*

Jul 2025 — Ayon Sarker, M.Sc. Engg.
*Design of Dual-band Plasmonic Absorber for Biomedical Sensing and Environmental Monitoring*

Jul 2025 — Soikot Sarkar, M.Sc. Engg.
*Hybrid Metal-Dielectric Nanostructures Integrated Heterojunction Thin Film Solar Cell for Efficiency*

May 2025 — Md. Asif Hossain Bhuiyan, M.Sc. Engg.
*Polarization Insensitive Electrically Reconfigurable Metasurface for Metalensing at Near Infrared Waveband*

Jan 2025 — Md. Mahfuzul Haque, M.Sc. Engg.
*Design of Silicon-Carbide Based Single-Quantum-Well White LED*

Jun 2024 — Md. Ehsanul Karim, M.Sc. Engg.
*Phase Change Material Based Broadband Multifunctional Metasurface for the Visible Range*

May 2023 — Shamima Akter Mitu, M.Sc. Engg.
*Design of an All-Optical Plasmonic Modulator for Two Micrometer Waveband*

### Ongoing

since Oct 2024 — Mir Md. Aminuzzaman, Ph.D.

since Oct 2024 — Tanvir Ahmed, M.Sc. Engg.

since Apr 2024 — Zaheer Al Jamee; Md. Al Shahriar Shakil; Naved Sadat Yamin; Jayed Hasan Rakib; all M.Sc. Engg.

since Apr 2023 — Abu Md. Raihan, M.Sc. Engg.

since Apr 2021 — Purbayan Das, M.Sc. Engg.

B.Sc. thesis — Addrita Basak Nodi; Ramima Mumtarin; Gopal Gosh (B.Sc. Engg., 2021 intake).

## Professional Service and Leadership

**IEEE**

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|---|---|
| Apr 2022 – present | Vice Chair, IEEE Photonics Society Bangladesh Chapter |
| Mar 2021 – Apr 2022 | Founding Chair, IEEE Photonics Society Bangladesh Chapter |
| Jul 2021 – May 2022 | Secretary, IEEE Bangladesh Section |
| Mar 2020 – Apr 2022 | Chair, IEEE Young Professionals Bangladesh |
| Jan 2013 – Dec 2013 | Chair, IEEE Graduates of the Last Decade (GOLD) |
| Jan 2011 – Dec 2012 | Vice-Chair, IEEE Graduates of the Last Decade (GOLD) |
| Jan 2011 – Dec 2012 | Student Activities Coordinator, IEEE Bangladesh Section |
| Jan 2008 – Aug 2009 | Chair, IEEE BUET Student Branch |
| Jan 2007 – Dec 2008 | Treasurer, IEEE BUET Student Branch |

**Optica**

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| May 2022 – present | Founding President, Optica Bangladesh Section |
| Jul 2022 – present | Founding Moderator, BUET Optical and Photonics Society |
| Jun 2016 – May 2017 | Treasurer, OSA Purdue Chapter, USA |

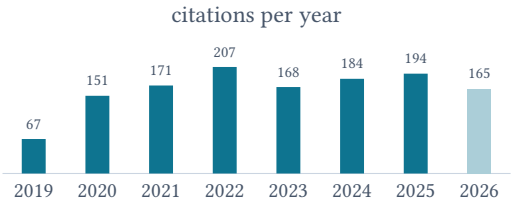
**Purdue University**

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| Jun 2017 – May 2018 | President, Nanotechnology Student Advisory Council (NSAC) |
| Jun 2016 – May 2017 | Vice-President, Nanotechnology Student Advisory Council (NSAC) |
| Jul 2017 – Jun 2018 | President, Bangladesh Students Association (Purdue BDSA) |
| Jul 2015 – Jun 2017 | Treasurer, Bangladesh Students Association (Purdue BDSA) |
| Jun 2015 – May 2016 | Treasurer, SPIE Purdue Chapter |

## Memberships

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|---|---|
| Apr 2021 – present | Member, National Young Academy of Bangladesh (NYAB) |
| | Senior Member, Institute of Electrical and Electronics Engineers (IEEE); Member, IEEE Photonics Society |
| | Member, Optica |
| Apr 2024 – present | Life Member, American Alumni Association of Bangladesh (AAAB) |
| Apr 2021 – present | Life Member, Association of BUET Alumni |

## Publications




**1,415** citations | **13** h-index | **17** i10-index

Google Scholar via Publish or Perish, retrieved 24 October 2026. 2026 is year to date.

Own name in **bold**; † student supervised. Numbered newest first within each category.

### Journal Articles

**J29** Subhan Zawad Bihan†, Anindya Kishore Choudhury and **Sajid Muhaimin Choudhury**. 'Steane [[7, 1, 3]] outer coding for loss-tolerant one-way quantum repeaters'. *Quantum Information Processing* 25.9 (Aug. 2026), p. 301.
doi:10.1007/s11128-026-05327-6 · SJR Q2

**J28** Dipika Rani Nath†, Sadid Muneer and **Sajid Muhaimin Choudhury**. 'Hybrid Si–GST polarization-insensitive dynamically tunable bifocal metalens operating at a 1.55μm wavelength'. *J. Opt. Soc. Am. B* 43.8 (Aug. 2026), B239–B250.
doi:10.1364/JOSAB.591582 · cited 1 · SJR Q2 · IF 1.9

**J27** Md. Mahfuzul Haque† and **Sajid Muhaimin Choudhury**. 'Tailoring electronic and optoelectronic properties of 2D-SiC via defects and doping: a first-principles study toward efficient white light-emitting diodes'. *RSC advances* 15 (2025), pp. 29335–29366.
doi:10.1039/D5RA04586J · cited 13 · SJR Q1 · IF 3.9

**J26** Md. Mahfuzul Haque†, Md. Rasidul Islam and **Sajid Muhaimin Choudhury**. 'Investigation of the physical properties through strain effect of monolayer silicon carbide material: DFT analysis'. *Physica B: Condensed Matter* 697 (2025), p. 416670.
doi:10.1016/j.physb.2024.416670 · cited 5 · SJR Q2 · IF 2.4

**J25** Md. Ehsanul Karim†, Md. Redwanul Karim and **Sajid Muhaimin Choudhury**. 'Synergizing deep learning and phase change materials for four-state broadband multifunctional metasurfaces in the visible range'. *Optics & Laser Technology* 181 (2025), p. 111730.
doi:10.1016/j.optlastec.2024.111730 · cited 7 · SJR Q1 · IF 5.2

**J24** Soikot Sarkar† and **Sajid Muhaimin Choudhury**. 'Efficiency enhancement of c-Si/TiO2 heterojunction thin film solar cell using hybrid metal-dielectric nanostructures'. *Solar Energy* 296 (2025), p. 113535.
doi:10.1016/j.solener.2025.113535 · cited 7 · SJR Q1 · IF 6.47

**J23** Ayon Sarker† and **Sajid Muhaimin Choudhury**. 'Concentric annular-hexagonal plasmonic resonator with nanorod vertices for dual-band absorption in NIR and MIR for sensing applications'. *Opt. Continuum* 4.5 (May 2025), pp. 1159–1173.
doi:10.1364/OPTCON.558501 · cited 1 · SJR Q3 · IF 1.1

**J22** Md. Asif Hossain Bhuiyan†, Purbayan Das† and **Sajid Muhaimin Choudhury**. 'Polarization insensitive electrically reconfigurable meta-lens for the 2 $\mu$m wavelength'. *Opt. Mater. Express* 14.12 (Dec. 2024), pp. 2830–2843.
doi:10.1364/OME.540435 · SJR Q2 · IF 2.8

**J21** Md. Ehsanul Karim† and **Sajid Muhaimin Choudhury**. 'Sb2S3/AlGaAs based Reconfigurable Metasurface for Dynamic Polarization and Directionality Control of Quantum Emitter Emission'. *RSC Advances* 40 (2024).
doi:10.1039/D4RA03726J · cited 3 · SJR Q1 · IF 3.9

**J20** Prithu Mahmud, Kaniz Fatema Supti and **Sajid Muhaimin Choudhury**. 'Lithium niobate photonic topological insulator-based multi-wavelength optical demultiplexer with piezoelectric switch-off'. *Opt. Express* 32.26 (Dec. 2024), pp. 45786–45800.
doi:10.1364/OE.541271 · cited 2 · SJR Q1 · IF 3.2

**J19** Mehedi Hasan Himel, Bejoy Sikder, Tanvir Ahmed and **Sajid Muhaimin Choudhury**. 'Biomimicry in Nanotechnology: A Comprehensive Review'. *NanoScale Advances* 5 (2023), pp. 595–614.
doi:10.1039/D2NA00571A · cited 54 · SJR Q1 · IF 5.6

**J18** Md. Ehsanul Karim† and **Sajid Muhaimin Choudhury**. 'Reconfigurable Broadband Metasurface with Switchable Functionalities in the Visible Range'. *Optical Materials Express* 13.5 (2023), pp. 1409–1423.
doi:10.1364/OME.489981 · cited 8 · SJR Q2 · IF 2.8

**J17** Md Asif Hossain Bhuiyan†, Shamima Akter Mitu† and **Sajid Muhaimin Choudhury**. 'TiN-GST-TiN all-optical reflection modulator for the 2$\mu$m wave band reaching 85% efficiency'. *Applied Optics* 61 (2022), pp. 9262–9270.
doi:10.1364/AO.470247 · cited 3 · SJR Q2 · IF 2.08

**J16** Himaddri Roy, Ehsanul Karim†, Sujoy Mondal and **Sajid Muhaimin Choudhury**. 'Custom Gold-Patterned Rewritable Optical Disc based Plasmonic Sensor for Blood Hemoglobin Detection'. *Optics Continuum* 1.10 (2022).
doi:10.1364/OPTCON.473106 · cited 1 · SJR Q3 · IF 1.1

**J15** Yasir Fatha Abed, Md Asif Hossain Bhuiyan† and **Sajid Muhaimin Choudhury**. 'T Grating on Nano-Cavity Array based Refractive Index Sensor'. *J. Opt. Soc. Am. B* 39.9 (2021).
doi:10.1364/JOSAB.426526 · cited 1 · SJR Q2 · IF 2.18

**J14** Mohammad Muntasir Hassan, Farhan Sadik Sium, Fariba Islam and **Sajid Muhaimin Choudhury**. 'A Review on Plasmonic and Metamaterial Based Biosensing Platforms for Virus Detection'. *Sensing and Bio-Sensing Research* 33 (2021), p. 100429.
doi:10.1016/j.sbsr.2021.100429 · cited 90 · SJR Q2 · IF 4.42

**J13** Ayon Sarker†, Shamima Akter Mitu†, Purbayan Das† and **Sajid Muhaimin Choudhury**. 'Structurally Tunable Gear-Shaped Plasmonic Sensor'. *Optics Express* 28.24 (2020), pp. 36070–36083.
doi:10.1364/OE.410123 · cited 14 · SJR Q1 · IF 3.67

**J12** Huan Jiang, **Sajid Muhaimin Choudhury**, Zhaxylyk A Kudyshev, Di Wang, Peng Xiao, Yongyuan Jiang and Alexander V Kildishev. 'Enhancing sensitivity to ambient refractive index with tunable few-layer graphene/hBN nanoribbons'. *Photonics Research* 7.7 (2019), pp. 815–822.
doi:10.1364/PRJ.7.000815 · cited 36 · SJR Q1 · IF 7.37

**J11** Huan Jiang, Harsha Reddy, Deesha Shah, Zhaxylyk A Kudyshev, **Sajid Muhaimin Choudhury**, Di Wang, Yongyuan Jiang and Alexander V Kildishev. 'Modulating Phase by Metasurfaces with Gated Ultra-thin TiN Films'. *Nanoscale* 11 (2019), pp. 11167–11172.
doi:10.1039/C9NR00205G · cited 10 · SJR Q1 · IF 6.62

**J10** Oscar Quevedo-Teruel, Hongsheng Chen, Ana Díaz-Rubio, Gurkan Gok, Anthony Grbic, Gabriele Minatti, Enrica Martini, Stefano Maci, George V Eleftheriades, Michael Chen, Nikolay I Zheludev, Nikitas Papasimakis, **Sajid Muhaimin Choudhury**, Zhaxylyk A Kudyshev, Soham Saha, Harsha Reddy, Alexandra Boltasseva, Vladimir M Shalaev, Alexander V Kildishev, Daniel Sievenpiper, Christophe Caloz, Andrea Alù, Qiong He, Lei Zhou, Guido Valerio, Eva Rajo-Iglesias, Zvonimir Sipus, Francisco Mesa, Raul Rodríguez-Berral, Francisco Medina, Victor Asadchy, Sergei Tretyakov and Christophe Craeye. 'Roadmap on metasurfaces'. *Journal of Optics* 21.7 (2019), p. 073002.
doi:10.1088/2040-8986/ab161d · cited 346 · SJR Q2 · IF 2.66

**J9** Maowen Song, Di Wang, Samuel Peana, **Sajid Muhaimin Choudhury**, Piotr Nyga, Zhaxylyk A. Kudyshev, Honglin Yu, Alexandra Boltasseva, Vladimir M. Shalaev and Alexander V. Kildishev. 'Colors with plasmonic nanostructures: A full-spectrum review'. *Applied Physics Reviews* 6 (2019), p. 041308.
doi:10.1063/1.5110051 · cited 264 · SJR Q1 · IF 16.30

**J8** **Sajid Muhaimin Choudhury**, Di Wang, Krishnakali Chaudhuri, Clayton DeVault, Alexander V Kildishev, Alexandra Boltasseva and Vladimir M Shalaev. 'Material platforms for optical metasurfaces'. *Nanophotonics* 7.6 (2018), pp. 959–987.
doi:10.1515/nanoph-2017-0130 · cited 196 · SJR Q1 · IF 8.09

**J7** **Sajid Muhaimin Choudhury**, Urcan Guler, Amr Shaltout, Vladimir M. Shalaev, Alexander V. Kildishev and Alexandra Boltasseva. 'Pancharatnam–Berry Phase Manipulating Metasurface for Visible Color Hologram Based on Low Loss Silver Thin Film'. *Advanced Optical Materials* 5 (2017), p. 1700196.
doi:10.1002/adom.201700196 · cited 88 · SJR Q1 · IF 8.70

**J6** Vladimir A. Zenin, **Sajid Muhaimin Choudhury**, Soham Saha, Vladimir M. Shalaev, Alexandra Boltasseva and Sergey I. Bozhevolnyi. 'Hybrid plasmonic waveguides formed by metal coating of dielectric ridges'. *Optics Express* 25.11 (2017), pp. 12295–12302.
doi:10.1364/OE.25.012295 · cited 40 · SJR Q1 · IF 3.67

**J5** Jongbum Kim, **Sajid Muhaimin Choudhury**, Clayton DeVault, Yang Zhao, Alexander V Kildishev, Vladimir M Shalaev, Andrea Alù and Alexandra Boltasseva. 'Controlling the Polarization State of Light with Plasmonic Metal Oxide Metasurface'. *ACS Nano* 10.10 (2016), pp. 9326–9333.
doi:10.1021/acsnano.6b03937 · cited 90 · SJR Q1 · IF 15.82

**J4** **Sajid Muhaimin Choudhury** and MA Matin. 'Multiport Analysis of Hexagonal Patch Antenna'. *IJECCT* 3.3 (2013).
https://journal.uniten.edu.my/index.php/ijecct/article/view/58 · cited 1

**J3** Md Gaffar, MA Zaman, **SM Choudhury** and Md A Matin. 'Design and optimisation of a novel dual-band circularly polarised microstrip antenna'. *IET Microwaves and Antennas & Propagation* 5.14 (2011), pp. 1670–1674.
doi:10.1049/iet-map.2010.0050 · cited 14 · SJR Q2 · IF 2.70

**J2** MA Zaman, SA Mamun, Md Gaffar, **SM Choudhury**, Md M Alam and Md Matin. 'Phased Array Synthesis Using Modified Particle Swarm Optimization'. *Journal of Engineering Science & Technology Review* 4.1 (2011).
doi:10.25103/jestr.041.10 · cited 19 · SJR Q3 · IF 1.6

**J1** Mohammad Asif Zaman, Md Gaffar, Md Mushfiqul Alam, Sayed Ashraf Mamun, **Sajid Muhaimin Choudhury** and MA Matin. 'Approximate Closed-Form Expression of the Electric Field of a Conical Horn Antenna'. *International Journal of Computer and Electrical Engineering* 3.1 (2011), p. 48.
https://ijcee.org/papers/291-E337.pdf · cited 1 · IF 1.6

## Conference Proceedings

**C27** **Sajid Muhaimin Choudhury**, Md. Ehsanul Karim†, Md. Redwanul Karim and Naved Sadat Yamin†. 'Integrating Deep Learning and Topology Optimization for Next-Generation Nanophotonic Devices'. In: *2025 Photonics & Electromagnetics Research Symposium - Spring (PIERS-Spring)*. 2025.
doi:10.1109/PIERS-Spring66516.2025.11276528

**C26** Md. Mahfuzul Haque† and **Sajid Muhaimin Choudhury**. 'Towards High-Performance Quantum-Based LEDs: Optoelectronic Modulation of 2D Silicon Carbide for Tunable and Efficient White Light Emission'. In: *2025 Photonics & Electromagnetics Research Symposium - Spring (PIERS-Spring)*. 2025.
doi:10.1109/PIERS-Spring66516.2025.11276753

**C25** Md Al Shahriar Shakil† and **Sajid Muhaimin Choudhury**. 'Half-Adder and Full-Adder Implementation with Continuous Variable Quantum Gates in a Photonic Quantum Computer'. In: *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*. 2025.
doi:10.1109/QPAIN66474.2025.11172078

**C24** Naved Sadat Yamin† and **Sajid Muhaimin Choudhury**. 'Valley Photonic Topological Insulator for Fluorescence Endoscopy'. In: *2025 Photonics & Electromagnetics Research Symposium - Spring (PIERS-Spring)*. 2025.
doi:10.1109/PIERS-Spring66516.2025.11276357

**C23** Mohammad Tahsin Alam, Yasir Mahmud, Zafrin Jahan Nikita and **Sajid Muhaimin Choudhury**. 'Gesture Controlled Bot with Temperature & Humidity (TH) Sensing Features'. In: *2024 2nd International Conference on Information and Communication Technology (ICICT)*. 2024, pp. 36–40.
doi:10.1109/ICICT64387.2024.10839649

**C22** Md Mahfuzul Haque† and **Sajid Muhaimin Choudhury**. 'DFT Analysis of Strain Effect in Bilayer 2D-SiC on Electrical, Optical, Phonon, and Thermodynamic Properties'. In: *2024 13th International Conference on Electrical and Computer Engineering (ICECE)*. 2024.
doi:10.1109/ICECE64886.2024.11024753 · cited 2

**C21** Md Asif Hossain Bhuiyan†, Shamima Akter Mitu† and **Sajid Muhaimin Choudhury**. 'VO2-based All-optical Reflection Modulator for 2$\mu$m Wave Band'. In: *2023 IEEE Photonics Conference (IPC)*. 2023, pp. 1–2.
doi:10.1109/IPC57732.2023.10360477 · cited 1

**C20** Saleh Ahmed Khan, Sadat Tahmeed Azad, Tiasa Mondal, Abdullah Jubair Bin Iqbal and **Sajid Muhaimin Choudhury**. 'Development of an Internet of Things based Bangla Calendar Clock'. In: *2023 26th International Conference on Computer and Information Technology (ICCIT)*. 2023, pp. 1–6.
doi:10.1109/ICCIT60459.2023.10441436

**C19** Abdul Mukit, Md Sabbir Hossen Bijoy, **Sajid Muhaimin Choudhury** and Md Tareq Mahmud. 'Discrete Modulated Continuous-Variable Quantum Key Distribution: Security and Noise Tolerance Enhanced by Decoy States and Effective Error Correction Protocol Integration'. In: *2023 IEEE International Conference on Telecommunications and Photonics (ICTP)*. 2023, pp. 1–5.
doi:10.1109/ICTP60248.2023.10490525 · cited 2

**C18** Kushol Roy Pritom, Md. Ehsanul Karim† and **Sajid Muhaimin Choudhury**. 'A Polarization Insensitive Achromatic Metalens Operating at Two Wavelengths in Visible Regime'. In: *2023 IEEE International Conference on Telecommunications and Photonics (ICTP)*. 2023, pp. 01–05.
doi:10.1109/ICTP60248.2023.10491019 · cited 1

**C17** Soikot Sarkar† and **Sajid Muhaimin Choudhury**. 'Design and Performance Analysis of a c-Si Thin-Film Solar Cell Using Plasmonic Ag Nanostructures'. In: *2023 IEEE International Conference on Telecommunications and Photonics (ICTP)*. 2023, pp. 01–05.
doi:10.1109/ICTP60248.2023.10490886

C16 Zhaxylyk A. Kudyshev, Ludmila J. Prokopeva, Maowen Song, **Sajid Muhaimin Choudhury** and Alexander V. Kildishev. 'Bi-anisotropic homogenization for efficient metasurface design (invited)'. In: *2018 International Applied Computational Electromagnetics Society Symposium (ACES)*. 2018, pp. 1–2.
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